

Calculation of DFT Spin–Orbit Spillage with Quantum ESPRESSO

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Abstract

This work describes the calculation of *spin–orbit spillage* from a crystal structure. Spin–orbit spillage provides a measure of the likelihood that a material has topological character. The spillage also provides the reference quantity for the machine-learning classifier of Choudhary et al. [4], which predicts whether the spillage exceeds a specified threshold rather than the calculation of its numerical value directly. The complete computation workflow was applied to the insulating compound BaMg₂Bi₂, yielding a spillage of 2.094 compared with the published VASP spillage of 2.075, corresponding to a difference of 0.9%. The calculation is described in terms of two Quantum ESPRESSO (QE) calculations of spillage performed with and without spin–orbit coupling, the role of relativistic pseudopotentials, and the subsequent wavefunction-overlap analysis. The limitations of the same calculation procedure for semimetals are also examined.

1 Physical overview

Band inversion of a material is an important indicator of its topological character [24]. Spin–orbit coupling (SOC), a relativistic effect that is particularly strong in heavy elements such as Bi, can change the band structure and occupation of electronic states. The **spillage** [1] quantifies the change in the occupied electronic bands upon inclusion of SOC. A small spillage indicates little change, whereas a large spillage can indicate band inversion and hence a candidate topological material. Choudhary et al. used this quantity to screen thousands of materials in the JARVIS database [2, 3].

Evaluation of this change of band structure requires electronic wavefunctions calculated *with* and *without* SOC. The computational workflow therefore consists of two density-functional-theory (DFT) calculations followed by a comparison of the corresponding occupied electronic bands.

The reference `jarvis-tools` implementation reads VASP wavefunction files. Here, the calculation is implemented directly from QE’s per-**k**-point HDF5 wavefunction files, providing a reusable QE workflow and allowing the effects of pseudopotential choice and occupied-band ambiguity to be examined explicitly.

The representative system is BaMg₂Bi₂ (JARVIS identifier JVASP-4053), a five-atom hexagonal crystal ($P\bar{3}m1$, $a = b = 4.906$ Å, $c = 8.293$ Å). This compound is an **insulator**, with a finite band gap between the highest filled and the lowest empty state at every **k**-point, 0.455 eV with SOC off. The band gap is narrow once SOC is switched on, 0.040 eV direct (and 0.021 eV indirect in JARVIS), but the gap never closes. As Section 12 shows, that single property of non-zero gap is what makes the calculation well posed.

2 Computational workflow

The diagram below summarizes the computational workflow.

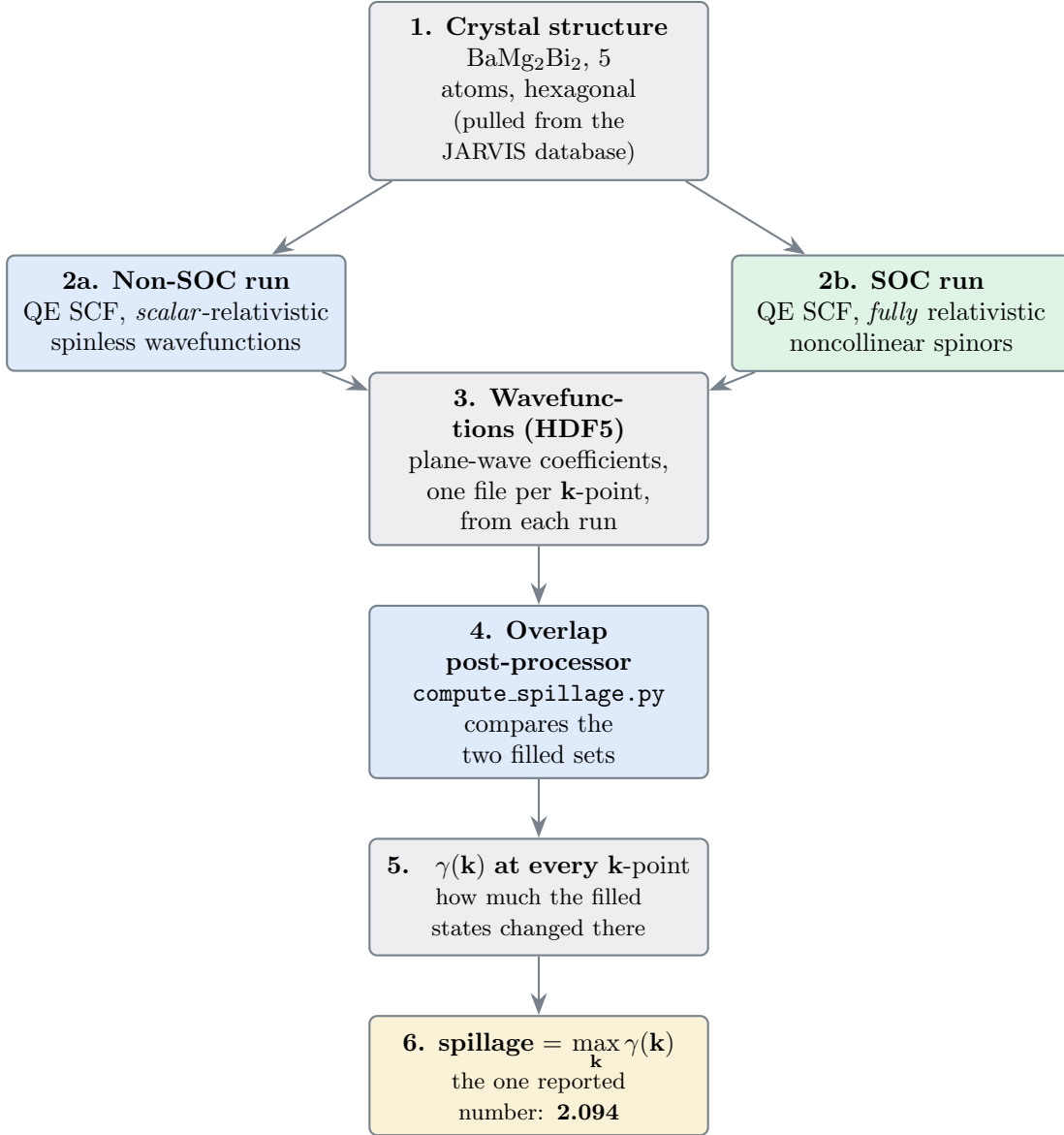


Figure 1: Workflow of the spillage computation.

The two DFT calculations dominate the computational cost. The choice of pseudopotentials for calculations 2a and 2b requires particular considerations, as discussed in Sections 6 and 7.

3 Definition of spin-orbit spillage

In the crystal’s momentum space, the “Brillouin zone,” DFT generates a list of electronic states (bands) at each **k**-point, arranged in order by energy. The lowest states are occupied and the rest are empty. Call the set of occupied states without SOC $\{|\psi_a^{\text{no}}(\mathbf{k})\rangle\}$ and with SOC $\{|\psi_m^{\text{so}}(\mathbf{k})\rangle\}$. There are N_{occ} occupied states in each.

The spillage at $\mathbf{k}$ is

$$\gamma(\mathbf{k}) = N_{\text{occ}} - \sum_{m=1}^{N_{\text{occ}}} \sum_{a=1}^{N_{\text{occ}}} |\langle \psi_m^{\text{so}}(\mathbf{k}) | \psi_a^{\text{no}}(\mathbf{k}) \rangle|^2. \quad (1)$$

The double sum measures the overlap between the occupied electronic states obtained by DFT with and without SOC. If SOC produces no change, every occupied state obtained with SOC aligns with the occupied state of the non-SOC calculation, the double sum equals the occupied-state count N_{occ} , and $\gamma(\mathbf{k}) = 0$. If SOC inverts bands so that occupied and unoccupied bands exchange in energy, the occupied states no longer coincide, the sum decreases below N_{occ} , and $\gamma(\mathbf{k}) > 0$.

The material's spillage is the maximum over all $\mathbf{k}$,

$$\eta = \max_{\mathbf{k}} \gamma(\mathbf{k}), \quad (2)$$

because a band inversion may occur at a particular crystal momentum. Usually the peak is at the $\mathbf{k}$ -point center $\Gamma = (0, 0, 0)$.

Assumption concerning band occupation. Equation (1) assumes a fixed and unambiguous value of N_{occ} at every $\mathbf{k}$. This condition is satisfied for an insulator but not necessarily for a metal or semimetal with zero band gap, as discussed in Section 12.

Required calculations. Equation (1) requires only the occupied DFT wavefunctions from calculations with and without SOC evaluated on the same $\mathbf{k}$ -point mesh. These quantities can be obtained using any plane-wave DFT code, e.g. VASP or QE.

4 DFT calculations with Quantum ESPRESSO (QE)

QE is an open-source software package [5, 6] that solves the Kohn–Sham equations of DFT [8, 7] for a crystalline material. Given a unit cell and atomic positions, QE computes the DFT electronic energies and wavefunctions of the material. The QE features relevant to the present calculation are summarized below.

Plane-wave basis.

The electronic wavefunction of the material is represented by plane waves. A plane wave is a wave whose value is constant on planes perpendicular to its wave direction.

Pseudopotentials.

Atomic core electrons near the nucleus are represented by spherical waves and are hard to represent with plane waves. To simplify the wave representation, the nucleus and core electrons are replaced by an effective potential called a pseudopotential, and only the valence electrons are treated explicitly. The choice of pseudopotential type is particularly important for the present calculation, as discussed in Sections 6 and 7.

The SCF loop.

The effective potential depends on the electron density being determined. The self-consistent-field (SCF) loop operates by iterating between solving for the wavefunctions and updating the density until convergence or self-consistency is achieved.

k-points.

Material properties are integrals over the Brillouin zone in reciprocal space, approximated by a finite Monkhorst–Pack [21] grid of $\mathbf{k}$ -points. A $6 \times 6 \times 4$ grid containing 144 points and centered at $(0, 0, 0)$ was used. The mesh is coarser along c because the cell is longer in that direction and its reciprocal axis is correspondingly shorter.

Relation between the cutoff and the Fourier representation. The plane-wave expansion is a Fourier series. A wavefunction in a crystal is periodic up to the Bloch phase, so

$$\psi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} \sum_{\mathbf{G}} c_{n\mathbf{k}}(\mathbf{G}) e^{i\mathbf{G}\cdot\mathbf{r}}, \quad (3)$$

where the complex coefficients $c_{n\mathbf{k}}(\mathbf{G})$ are exactly the Fourier components of the cell-periodic part and the $\mathbf{G}$ vectors are reciprocal-lattice vectors. The $\mathbf{G}$ -vector coefficients are used in the spillage calculations. The wavefunction cutoff truncates the Fourier series, and `ecutwfc` defines this truncation. A bigger cutoff means more plane waves and greater accuracy but higher cost.

Two aspects of the cutoff definition require clarification.

- `ecutwfc = 70` means “keep every plane wave whose kinetic energy $\hbar^2|\mathbf{k} + \mathbf{G}|^2/2m$ is below 70 Ry.” In Rydberg atomic units, the condition reduces to $|\mathbf{k} + \mathbf{G}|^2 \leq 70$, i.e. the $\mathbf{G}$ vectors inside a *sphere* of radius $\sqrt{70} = 8.37$ bohr⁻¹ in reciprocal space. For BaMg₂Bi₂ that sphere holds 11,573 plane waves at Γ . Section 8 derives an estimate of this plane-wave count.
- **ecutrho is what sets the fast Fourier transform (FFT).** The SCF loop moves between real and reciprocal space every iteration, and the loop is implemented with an FFT on a uniform grid. Reciprocal space is the Fourier transform of real space. The electron density is the square of the wavefunctions, so its reciprocal components need to extend to twice the wavefunction cutoff. Because energy is proportional to the square of the reciprocal-vector magnitude, this requires `ecutrho = 4 × ecutwfc = 280` Ry for the norm-conserving default. That cutoff fixes the FFT box. QE reports a $50 \times 50 \times 90$ grid holding 92,297 density reciprocal-lattice $\mathbf{G}$ -vectors for this cell. The 90 is along c because the cell is longest there and therefore needs the most sample points. Section 8 derives an estimate of this count.

5 Calculations with and without spin–orbit coupling

The spillage requires two SCF calculations of the same crystal structure, one with SOC and one without SOC. The calculations differ only in the SOC calculation control parameters and the corresponding pseudopotentials, as summarized in Table 1. The atomic structures were obtained from the JARVIS database after relaxation with OptB88vdW, and the present SCF calculations use PBE; see Listing 1. This matches the procedure of Choudhary et al. [2], who evaluated PBE spillage on OptB88vdW-relaxed structures because OptB88vdW with SOC was not available in VASP.

`noncolin=.true.` makes every wavefunction a two-component *spinor* (a spin-up part and a spin-down part, each its own list of plane-wave coefficients). `lspinorb=.true.` adds the actual spin–orbit interaction. Together they are “the SOC calculation.”

Band counts. The difference between 48 and 96 bands arises from spin degeneracy rather than a different convergence choice. Without SOC, every spatial orbital holds one spin-up and one spin-down electron, so 60 electrons need only 30 bands and `nbnd = 48` leaves a 60% margin of 18 empty

Table 1: The two SCF calculations. All control parameters are identical except the two SOC switches and the pseudopotential type required. Calculations 2a and 2b correspond to Figure 1.

	Non-SOC run (2a)	SOC run (2b)
<code>noncolin</code>	<code>.false.</code> (spin-degenerate)	<code>.true.</code> (spinor wavefunctions)
<code>lspinorb</code>	<code>.false.</code> (SOC off)	<code>.true.</code> (spin-orbit term on)
pseudopotentials	scalar-relativistic (<code>_SR</code>)	fully relativistic (<code>_FR</code>)
complex output	one list	<i>two</i> lists (spin up, spin down)
bands computed (<code>nbnd</code>)	48	96

bands. With SOC the ordinary spin-up/spin-down occupation convention is gone: QE considers each one-electron spinor separately, so 60 electrons need 60 bands. Kramers pairs can still remain degenerate. Choosing `nbnd`= 96 leaves 36 empty bands, the same 60% fractional margin. Both runs therefore describe the same 60 electrons, although the SOC calculation requires twice as many band labels. This same factor of two reappears in the post-processor as $N_{\text{occ}}/2$ versus N_{occ} in Section 9.

Consistency between the two calculations. The cell, the `ecutwfc/ecutrho` values, and the **k**-point mesh are identical. The settings `nosym=.true.` and `noinv=.true.` turn off symmetry reduction and time-reversal folding applied by QE. Symmetry would let QE compute fewer **k**-points and reorder plane waves, but SOC lowers the symmetry differently from the non-SOC run, so the two runs would end up with mismatched **k**-point lists and plane-wave orderings. With symmetry off, both runs use the *same* full **k**-grid and the *same* plane-wave set, so the wavefunctions line up one-to-one and the overlap in Eq. (1) is well defined in the two calculations.

Listing 1: Relevant parameters in the SOC input file for BaMg2Bi2.

```
&system
  ecutwfc = 70.0      ! plane-wave cutoff (Ry); MUST match the non-SOC run
  ecutrho = 280.0     ! density cutoff (Ry); MUST match the non-SOC run
  nbnd    = 96        ! how many bands to compute (>= occupied + margin)
  noncolin = .true.    ! <-- spinor wavefunctions
  lspinorb = .true.    ! <-- spin-orbit coupling ON (this is what 2a lacks)
  nosym    = .true.    ! no symmetry reduction, so k-points/G-vectors match run 2a
  noinv    = .true.    ! ... and no time-reversal folding either
  occupations = 'smearing'
  smearing = 'mp'      ! Methfessel-Paxton (mp)
  degauss  = 0.01     ! smearing width in Ry (see the caveat on occupations)
/
ATOMIC_SPECIES
  Ba 137.327 Ba_FR.upf ! <-- "_FR" = fully relativistic. This is where SOC lives.
  Mg  24.305 Mg_FR.upf
  Bi 208.980 Bi_FR.upf
```

Fields in ATOMIC_SPECIES (Ba, Mg, Bi). The three fields specify the atomic species label, atomic mass, and pseudopotential filename.

- The **atomic label** (Ba) identifies the atomic species in the `ATOMIC_POSITIONS` block. The label is not used for an external chemical lookup. Distinct labels may therefore be assigned to inequivalent sites of the same element when different pseudopotentials are required.

- The **mass** (137.327) is the isotope-weighted atomic mass in atomic mass units. It enters calculations involving nuclear inertia or mass weighting, such as molecular dynamics and vibrational analysis. The mass does not affect the single-point SCF calculations currently performed at fixed atomic positions.
- The **filename** (Ba_FR.upf) is read as specified from **pseudo_dir**. That file defines the element identity. The file header contains the element symbol, the valence charge Z_{val} , the exchange–correlation functional for which the atomic species was built, and, for an _FR file, the j -resolved channels that carry SOC.

For instance, the specified .upf file identifies the element and valence charge (for example **z_valence**="15.00" for Bi), while the numerical tables in the file encode the ionic potential and valence configuration.

6 Relativistic treatment

6.1 Relativistic effects in electronic-structure calculations

Electrons bound to a heavy nucleus can move very fast, at an appreciable fraction of the speed of light, calling for consideration of special relativity. Near a Bi nucleus ($Z = 83$), relativistic effects are therefore significant. The single-electron description formulated by the Schrödinger equation is no longer adequate and should be replaced by the Dirac equation [10], which represents the electron using a four-component wavefunction.

Expansion of the Dirac equation in powers of $1/c$ yields three corrections to the Schrödinger description. The mass–velocity term (H_{mv}) and Darwin term (H_{D}) shift the energies without coupling to spin. The third correction is the **spin–orbit-coupling term** (H_{SO}),

$$H_{\text{SO}} \propto \frac{1}{r} \frac{dV}{dr} \mathbf{L} \cdot \mathbf{S}, \quad (4)$$

which couples the spin $\mathbf{S}$ to the orbital angular momentum $\mathbf{L}$ of the electron. The H_{SO} term can result in electronic band inversion and topological electronic states. The H_{SO} magnitude increases strongly with nuclear charge, making Bi-containing compounds prominent among candidate topological materials.

6.2 Scalar- and fully relativistic pseudopotentials

The distinction between “scalar-” and “fully-” relativistic treatments differs by keeping the first two correction terms or all three terms, respectively.

Scalar-relativistic (SR).

Keeps the mass–velocity and Darwin terms, and *averages* H_{SO} away. Specifically, the atomic solve is done in the Koelling–Harmon scheme [11], which discards the spin–orbit operator and produces states labelled by electronic orbital angular momentum ℓ alone. The result includes relativistic energy corrections but excludes spin–orbit coupling. Calculation 2a therefore uses Ba_SR.upf.

Fully-relativistic (FR).

Solves the Dirac equation properly, so states are labelled by the *total* electronic angular momentum $j = \ell \pm \frac{1}{2}$. A p orbital is no longer one object; it splits into $p_{1/2}$ and $p_{3/2}$ with different radial potentials, and the size of that splitting is the spin–orbit coupling strength. Storing both j channels is how a .upf file carries SOC [13, 12]. Calculation 2b therefore uses Ba_FR.upf.

When `lspinorb=.true.`, QE reads those j -resolved channels and builds the spin-orbit operator from the difference between them; the noncollinear/spinor machinery that makes this difference possible in a plane-wave code is described in Ref. [14]. With `lspinorb=.false.` QE would average the two j channels. An FR pseudopotential and the corresponding SOC flag are both required.

6.3 Consequences for the spillage calculation

Invariance of the plane-wave basis. The plane-wave set is fixed by the cutoff, the cell, and the $\mathbf{k}$ -points and is independent of the pseudopotential. So swapping SR for FR leaves the basis unchanged. The two calculations therefore remain directly comparable, and their difference isolates the effect of SOC. This unchanged basis is verified numerically in Section 8.

Spinor representation. Without SOC, each spatial orbital holds one spin-up and one spin-down electron; QE stores one list of coefficients per band. With SOC, spin and orbital angular momenta are entangled, no electronic state is purely up or down, and QE stores *two* lists per band. That doubling is the only structural change the post-processor has to accommodate to compute the spillage.

6.4 Post-processing

Relativistic effects are incorporated before post-processing. The PseudoDojo pseudopotentials used here [16, 15] were generated by solving the Dirac equation for each isolated atom and storing the result in j -resolved channels. QE reads these channels and, when `lspinorb=.true.`, assembles the spin-orbit operator during the SCF cycle. The resulting wavefunctions therefore already contain the relativistic contributions.

`compute_spillage.py` performs linear-algebra operations on the output using Equations (1) and (6); see Listing 2.

Listing 2: Treatment of SOC and non-SOC wavefunction arrays in the post-processor. `igwx` is the number of plane waves at this $\mathbf{k}$ -point (11,573 at Γ).

```
# read_wfc(), for EITHER run: the stored array has npol blocks of igwx columns
npol = int(a["npol"])          # 2 for the SOC run, 1 for the non-SOC run
c    = ...                    # shape (nbnd, npol * igwx)

# --- non-SOC run: npol = 1, so there is ONE block and nothing to split ---
Cns = ns["c"][:N_OCC_SPATIAL]  # (N_occ/2, igwx)

# --- SOC run: npol = 2, so the row is sliced into its two halves ---
up = soc["c"][:, :soc["igwx"][:, perm] # (nbnd, igwx) spin-up block
dn = soc["c"][:, soc["igwx"]*2 * soc["igwx"][:, perm] # (nbnd, igwx) spin-down block
```

Representation of the non-SOC wavefunctions. The relevant statement is the `Cns` assignment. Both calculations store a band as a row of $\text{npol} \times \text{igwx}$ complex numbers. With `npol = 1`, that row is the single spinless coefficient vector and requires no additional slicing. With `npol = 2`, the row is twice as long and is separated at `igwx` into spin-up and spin-down components. Thus, `npol` determines whether this slicing is required.

Separating the stored array into two components does not itself constitute a relativistic calculation. Equation (4) is incorporated by the pseudopotential generator, activated in the QE input, and represented in the wavefunctions read by the post-processing code.

7 Choice of pseudopotential and overlap metric

Beyond SR-vs-FR there is a second, independent choice of pseudopotentials, and the choice decides which overlap metric must be used in Eq. (1).

- With **norm-conserving (NC)** pseudopotentials, a wavefunction is represented by its plane-wave coefficients, and $\langle \psi | \psi' \rangle$ is the dot product of those two lists. That dot product is the correct overlap within the norm-conserving pseudo-wavefunction representation.
- With **PAW** [18] (or ultrasoft) pseudopotentials, the stored wavefunction is a smoothed version that is missing a piece near each atomic core. The true overlap needs an extra augmentation correction, which is not contained in the wavefunction file. Omitting this correction is invalid because even $\langle \psi | \psi \rangle$ is not 1.

A calculation using fully relativistic **PAW** pseudopotentials was initially conducted on Ba₃BiSb (see Section 12). The resulting overlap produced an unphysical raw value of $\gamma \approx -17$, likely caused by a poor augmentation correction. Re-orthonormalization of the occupied states using a Löwdin correction [23] approximated the missing augmentation and yielded 1.63, but introduced an uncontrolled approximation.

This normalization issue was resolved by using **norm-conserving** FR pseudopotentials (PseudoDojo NC v0.4 [16, 15], PBE functional [9]), so that the ordinary coefficient dot product is the appropriate overlap and no PAW augmentation correction is needed. All subsequent results use NC pseudopotentials. For BaMg₂Bi₂, the valence counts are Ba 10 ($5s^2 5p^6 6s^2$), Mg 10 ($2s^2 2p^6 3s^2$), and Bi 15 ($5d^{10} 6s^2 6p^3$), giving $10 + 2(10) + 2(15) = 60$.

8 Wavefunction output

After each SCF calculation, QE writes the wavefunctions to HDF5 files, one per **k**-point (`wfc1.hdf5`, `wfc2.hdf5`, ...). Two wavefunction datasets are used in the overlap analysis.

- **MillerIndices**: the list of **G** vectors (which plane waves are present at this **k**).
- **evc**: the coefficients, one complex number per plane wave per band. In the SOC run each band has *two* blocks of coefficients (spin up, then spin down), because the wavefunction is a spinor.

The common basis of the SR and FR calculations can be verified from these files. At Γ , both BaMg₂Bi₂ plane-wave calculations report exactly 11573 plane waves. This agreement follows from the common cutoff rather than the pseudopotential type and permits direct overlap of the **G**-vector coefficients without interpolation.

Estimate of the plane-wave count. The expected plane-wave count N_{PW} can be estimated directly from the cutoff. The count is independent of the 144 **k**-points and 60 electrons because each **k**-point has a basis of approximately this size, and the electron count determines only the number of occupied bands. The plane-wave count is set by the cell and cutoff.

Counting **G** vectors is counting reciprocal-lattice points inside a sphere. Reciprocal space has one lattice point per volume $(2\pi)^3/V$, where V is the real-space cell volume, and the cutoff keeps all reciprocal-lattice points inside a sphere of radius $G_{\max}$ using the wavefunction cutoff `ecutwfc`, where $G_{\max} = \sqrt{\text{ecutwfc}}$ in bohr⁻¹. Thus

$$N_{PW} \approx \frac{\frac{4}{3}\pi G_{\max}^3}{(2\pi)^3/V} = \frac{V G_{\max}^3}{6\pi^2} = \frac{V (\text{ecutwfc})^{3/2}}{6\pi^2}. \quad (5)$$

QE prints the cell volume of BaMg₂Bi₂ as $V = 1166.38 \text{ bohr}^3$, and $G_{\text{max}} = \sqrt{70} = 8.367 \text{ bohr}^{-1}$, so

$$N_{\text{PW}} \approx \frac{1166.38 \times 585.7}{59.22} = 11,535,$$

against QE's 11,573, which is 0.3% low. The small shortfall of N_{PW} is expected. A sphere cut out of a discrete lattice never contains exactly the continuum estimate, and the discrepancy shrinks as the sphere grows. Running Equation (5) using the density cutoff `ecutrho` is the sharper test, because that sphere is eight times larger: `ecutrho` = 280 Ry gives $G_{\text{max}} = 16.733 \text{ bohr}^{-1}$ and predicts 92,284, against the 92,297 **G**-vectors QE reports for the density grid. That is an error of 0.01%.

9 Wavefunction-overlap post-processing

The script `compute_spillage.py` implements Eq. (1); see Listing 3. For each **k**-point the code does the following:

1. **Read** the non-SOC and SOC wavefunction files for that **k**.
2. **Align** the plane waves by their Miller indices so that coefficient g in one calculation aligns one-to-one with the same **G** as coefficient g in the other. The correspondence as controlled by `nosym` is verified explicitly.
3. **Take the filled states:** the lowest $N_{\text{occ}}/2$ spatial orbitals from the non-SOC run and the lowest N_{occ} spinor states from the SOC run. (N_{occ} = number of valence electrons, 60 for BaMg₂Bi₂. The non-SOC run is spin-degenerate, so each of its 30 orbitals stands for a spin-up and a spin-down state, giving 60 filled electronic states to match the 60 SOC ones.)
4. **Build each non-SOC orbital as a spinor:** φ_a becomes $|\varphi_a, \uparrow\rangle$ and $|\varphi_a, \downarrow\rangle$. Then the overlap of a SOC spinor $|\psi_m^{\text{so}}\rangle = (u_m, d_m)$ with these states is given by $\langle u_m | \varphi_a \rangle$ and $\langle d_m | \varphi_a \rangle$.
5. **Sum** the squared overlaps and subtract from the occupied-state count N_{occ} to give γ :

$$\gamma(\mathbf{k}) = N_{\text{occ}} - \sum_{m=1}^{N_{\text{occ}}} \sum_{a=1}^{N_{\text{occ}}/2} \left(|\langle u_m | \varphi_a \rangle|^2 + |\langle d_m | \varphi_a \rangle|^2 \right). \quad (6)$$

These overlaps are plugged into Equation (1), resulting in Equation (6). Finally the script takes $\max_{\mathbf{k}} \gamma(\mathbf{k})$, which is the material spillage. The script prints two columns, `gamma_raw` (dot products) and `gamma_lowdin` (after the re-orthonormalization correction). With NC pseudopotentials the dot product is the appropriate pseudo-wavefunction overlap. The correction was needed when PAW pseudopotentials were used. The two columns must therefore agree apart from numerical error. Their agreement to four decimal places is used as a regression test of the implementation rather than as evidence for the overlap metric.

Listing 3: Calculation at each **k**-point (from `compute_spillage.py`). Every array is (rows = electronic states) $\times$ (columns = plane waves). For BaMg₂Bi₂ at Γ : $N_{\text{occ}} = 60$, so the array dimensions are (30, 11573) and (60, 11573).

```
Cns      = nonSOC occupied orbitals      # (N_occ/2, n_planewaves)  <- 30 rows, 11573
          columns
up, dn = SOC occupied spinors            # each (N_occ, n_planewaves) <- 60 rows, 11573
          columns
```

```

A_up   = up.conj() @ Cns.T           # dot product (N_occ, N_occ/2) <u_m | phi_a>
A_dn   = dn.conj() @ Cns.T           # dot product (N_occ, N_occ/2) <d_m | phi_a>
gamma   = N_occ - ( |A_up|^2 + |A_dn|^2 ).sum()

```

Electronic-state and plane-wave counts. The first dimension of each array is the number of electronic states, whereas the second is the number of plane waves. Only the first dimension is halved. The plane-wave dimension is 11,573 in both calculations. If the two calculations did not share a common basis, the matrix product `up.conj() @ Cns.T` would not even be conformable.

The halving is a spin-counting statement. The non-SOC run is spin-degenerate, so QE stores each orbital once and lets it hold two electrons; 60 electrons therefore occupy $N_{\text{occ}}/2 = 30$ orbitals. The SOC run has no such degeneracy, so QE stores all 60 occupied spinor states separately. Step 5 in Listing 3 reconciles the two: each of the 30 orbitals is used twice, once against the spin-up half of a spinor and once against the spin-down half, which is why there are two overlap matrices `A_up` and `A_dn` of dimension (60, 30) rather than one of dimension (60, 60). Together they carry the same 60×60 overlaps required by Equation (1) or Equation (6).

Note also that the script fixes $N_{\text{occ}} = 60$ for every $\mathbf{k}$ -point rather than reading QE’s occupation numbers. Section 11 justifies that choice for BaMg_2Bi_2 , and Section 12 discusses the limitation of semimetals.

10 Result for BaMg_2Bi_2

Table 2: QE results obtained for BaMg_2Bi_2 (JVASP-4053). The values are compared with published JARVIS VASP values. The spillage reference is from `dft_3d`; the two band-gap references are from the TopoMoat table.

quantity	QE (this work)	JARVIS (VASP)	difference
spillage $\eta = \max_{\mathbf{k}} \gamma$	2.094	2.075	+0.9%
direct gap, no SOC	0.455 eV	0.451 eV	+0.9%
direct gap, with SOC	0.040 eV	0.040 eV	same at shown precision

The calculations used NC pseudopotentials (SR and FR), a $6 \times 6 \times 4$ mesh containing 144 $\mathbf{k}$ -points, 60 electrons, and `ecutwfc` = 70 Ry. On an Apple M2 Pro processor, the non-SOC and SOC SCF calculations took 2.3 and 12.7 minutes, respectively. The time needed for post-processing to yield 2.094 is short.

Analysis of the result.

- **The overlap implementation passes a regression test.** For NC pseudopotentials, the overlap operator is the identity, $\langle \phi | \phi \rangle = 1$, and the QE eigenvectors are already orthonormal under the coefficient dot product. Equality of `gamma_raw` and `gamma_lowdin` is therefore expected analytically. Their agreement at all 144 $\mathbf{k}$ -points to four decimal places confirms the plane-wave alignment and array handling, but does not independently validate the overlap metric or remove the usual finite-cutoff, finite-mesh, and pseudopotential approximations.
- **The band structure agrees independently.** The two band-gap results above are not part of the spillage; they independently verify that the QE electronic structure agrees with the VASP

result that produced the reference. They agree to 1% or better, which gives confidence in the 0.9% spillage agreement. Note also what the two rows say between them: switching SOC on nearly closes a half-electron-volt gap at Γ , which is exactly where γ peaks. That is the same reshuffling of the filled states the spillage quantifies, seen in the band energies instead.

- **The sampled peak is sharply localized at Γ .** The spillage reaches 2.094 at Γ and falls to 0.22 at the nearest sampled $\mathbf{k}$ -points. Because Γ remains present on a refined Γ -centred mesh, refinement cannot lower this sampled value of 2.094. It could reveal a larger value at an intermediate point, however, so formal $\mathbf{k}$ -mesh convergence still requires a denser calculation.

Reference value. JARVIS reports two DFT spillages for JVASP-4053, namely 2.075 in the `dft_3d` table [2] (the one the ML screen is trained on and the one quoted above) and 2.081 in the TopoMoat table [3]. The present value of 2.094 is 0.9% above the first and 0.6% above the second, so the conclusion does not depend on which is taken as canonical.

11 Validation of the occupied-state count

The post-processing calculation of the spillage of BaMg_2Bi_2 fixes the occupied-state count $N_{\text{occ}} = 60$ throughout the Brillouin zone. This choice was validated by direct examination of the DFT band energies at all 144 $\mathbf{k}$ -points in both calculations.

Table 3: The band gap above the 60th filled state never closes or becomes zero anywhere in the Brillouin zone in either run.

	non-SOC ($E_{31} - E_{30}$)	SOC ($E_{61} - E_{60}$)
smallest gap over all $\mathbf{k}$	0.455 eV (at Γ)	0.040 eV (at Γ)
$\mathbf{k}$ -points where the gap closes	0 of 144	0 of 144

As shown in Table 3, a finite non-zero gap separates the 60th and 61st states throughout the Brillouin zone in both calculations. “The lowest 60” is therefore unambiguous at every $\mathbf{k}$ -point, and $N_{\text{occ}} = 60$ is exact, not an approximation.

Effect of smearing on the reported occupations. The input files use Methfessel–Paxton smearing [22] with width defined by `degauss` = 0.01 Ry = 0.136 eV, inherited from the earlier semimetal calculations. This width exceeds the 0.040 eV SOC gap, so the QE occupation numbers at Γ come out fractional even though a genuine gap is present, i.e. $f = 1$ below the gap in the ideal limit. This does *not* affect the reported spillage value because the post-processor does not use these occupations and instead uses the fixed count justified above. Table 4 reports the QE occupations near the band boundary and compares them with the values for an ideal insulator.

The table reflects two distinct effects of the occupations.

- **The values above 1 and below 0 result from the smearing scheme, not the physics.** Methfessel–Paxton (MP) is not a softened step function. It is a Gaussian function multiplied by Hermite-polynomial corrections, chosen so that corrected quantities in DFT calculations converge quickly with smearing width. The price is that the occupation number oscillates slightly outside $[0, 1]$ near the Fermi level. Thus 1.0126 and -0.0279 in the non-SOC run are the expected signature of MP working as designed. They are not evidence of a metal: that run

Table 4: QE occupation numbers f at Γ for BaMg₂Bi₂ near the occupied–unoccupied boundary, compared with the ideal values of 1 and 0 for an insulator. Both calculations report occupations on a 0–1 scale; in the non-SOC calculation the factor of two for spin is included in the $\mathbf{k}$ -point weight, not here. Fermi levels: 7.489 eV (non-SOC), 7.835 eV (SOC).

calculation	bands	E (eV)	reported f	ideal f	interpretation
non-SOC	1–27	≤ 6.220	1.0000	1	ideal occupation
	28	7.177	1.0027	1	MP overshoot
	29, 30	7.238	1.0126	1	MP overshoot; 30 is the last filled
	31	7.693	−0.0279	0	MP undershoot; first empty
	32–48	≥ 8.141	−0.0000	0	non-ideal occupation
SOC	1–56	≤ 6.643	1.0000	1	ideal occupation
	57, 58	7.729	0.9838	1	leakage
	59, 60	7.798	0.7218	1	leakage; 60 is the last filled
	61, 62	7.838	0.4826	0	first empty; the gap here is 0.040 eV
	63, 64	8.153	−0.0023	0	MP undershoot
	65–96	≥ 9.157	−0.0000	0	non-ideal occupation

has a 0.455 eV gap, more than three times the smearing width. A Gaussian or Fermi–Dirac smearing would have stayed inside $[0, 1]$ and converged more slowly.

- **The 0.72 and 0.48 values in the SOC calculation reflect charge leakage.** The gap is 0.040 eV and the smearing width is 0.136 eV, so the smearing genuinely cannot resolve the gap and moves charge across it. The reported values may therefore be misleading. Interpreted directly, band 61 appears half-occupied, and a criterion based on $f \geq 0.5$ would count 62 states as filled rather than 60. Section 12 shows the resulting discrepancy on materials where the count is genuinely ambiguous.

These f values are not required to sum to the occupied-state count at an individual $\mathbf{k}$ -point. At Γ , the non-SOC values sum to 30.0000, whereas the SOC values sum to approximately 60.37. Only the Brillouin-zone-weighted total is constrained. These f values do not enter the spillage because N_{occ} is fixed at 60 according to Table 3. Nevertheless, using `occupations='fixed'` or a `degauss` value substantially below the band gap would be more appropriate for an insulating system.

12 Limitations for semimetals

The computational workflow was initially applied using PAW pseudopotentials (PP) to the cubic semimetal Ba₃BiSb (JVASP-36485) and showed substantially poorer agreement with the reference value. This discrepancy arises from a limitation of the spillage definition for semimetals rather than necessarily from the numerical implementation of the spillage. The discrepancy is larger than for other materials, as shown in Table 5.

The comparison across material classes also contains a $\mathbf{k}$ -mesh-size difference. The Ba₃BiSb semimetal was evaluated on a $4 \times 4 \times 4$ mesh, whereas the three reference insulators used $6 \times 6 \times 4$ or $6 \times 6 \times 6$ meshes. Because semimetallic states near the Fermi level can be particularly sensitive to Brillouin-zone sampling, part of the 11% discrepancy may reflect this difference, and the comparison does not isolate the occupied-band ambiguity alone.

Table 5: Calculated spillage values η and comparison with published references.

material	type	PP type	k -mesh	e^-	η	reference
BaMg ₂ Bi ₂ (JVASP-4053)	insulator	SR/FR NC	$6 \times 6 \times 4$	60	2.094	2.075 (+0.9%)
Bi ₂ Se ₃ (JVASP-1067)	insulator	SR/FR NC	$6 \times 6 \times 6$	78	2.115	2.098 (+0.8%)
Bi ₂ Te ₃ (JVASP-25)	insulator	SR/FR NC	$6 \times 6 \times 6$	78	2.112	2.094 (+0.8%)
Ba ₃ BiSb (JVASP-36485)	semimetal	SR/FR NC	$4 \times 4 \times 4$	60	2.016	2.267 (−11%)
Ba ₃ BiSb, initial calculation	semimetal	SR/FR PAW	$2 \times 2 \times 2$	50	1.632	2.267 (−28%)
PbTe (JVASP-1103)	insulator	SR/FR NC	$6 \times 6 \times 6$	30	2.010	2.200 [28]

Agreement for clean-gap insulators. The values calculated for BaMg₂Bi₂, Bi₂Se₃, and Bi₂Te₃ exceed the corresponding published VASP values by 0.9%, 0.8%, and 0.8%, respectively. The consistent sign and magnitude suggest a small systematic offset between the QE norm-conserving and VASP PAW calculations rather than statistical variation. An offset below one percent is consistent with the expected agreement between converged plane-wave codes [26]. The −11% discrepancy for the semimetal is therefore qualitatively distinct. All three insulating systems also satisfy the condition of Section 11: the gap above the last filled band never closes anywhere in the zone (0 of 144 **k**-points for BaMg₂Bi₂, 0 of 216 for each of the other two).

The Bi₂X₃ calculations ($X = \text{Se or Te}$) used 216 **k**-points and occupied-state counts of 78. Calculations based on fixed counts of 78 reproduced 2.1146 and 2.1116 to four decimal places, confirming that the fixed- N_{occ} post-processing is insensitive to the occupations reported by QE.

Origin of the discrepancy. Ba₃BiSb is a semimetal with an indirect gap of −0.178 eV, indicating that the occupied and empty bands overlap in energy. At Γ , which is exactly where its spillage peaks, a *triply degenerate* state lies at the Fermi level. The occupied subspace required by Eq. (1) is therefore ill-defined at this point. Shifting the occupation boundary by two bands produces substantial changes in the calculated value; see Table 6.

 Table 6: Ba₃BiSb: the same wavefunctions and the same code, with only the assumed occupied count changed.

electrons assumed occupied	56	58	60	62	64
resulting $\gamma(\Gamma)$ for Ba ₃ BiSb	2.12	1.43	2.02	0.08	1.63

For identical wavefunctions and code, changing the assumed occupied-state count by two bands changes γ from 0.08 to 2.12. The related compound Ba₃Bi₂ (JVASP-36513) exhibits an even larger sensitivity, as shown in Table 7.

The `jarvis-tools` criterion counts bands with occupation ≥ 0.5 separately at each **k** rather than fixing the count. On Ba₃Bi₂, the script selects 62 electrons and yields a spillage of 4.12 compared with the reference value of 4.097, a difference of 0.5%. Applied unchanged to Ba₃BiSb, the script also selects 62 electrons but yields 0.08 compared with the reference value of 2.267. The value of 4.12 is therefore reported as a diagnostic comparison rather than a validated workflow result. The two materials agree closely at 56 and 58 electrons and only diverge once the count enters the degenerate manifold at E_F , which is what identifies the manifold as the source of the instability. Here `jarvis-tools` refers to the separate JARVIS Python package, not to a component of QE or the QE conda-forge build.

Table 7: Ba_3Bi_2 (JVASP-36513) against Ba_3BiSb (JVASP-36485), $\gamma(\Gamma)$ as a function of the assumed occupied-state count. Both are antiperovskite semimetals with a degenerate manifold on E_F at Γ . The values span 0.08 to about 4.2; the `jarvis-tools` rule selects 62 electrons and gives 4.12 for Ba_3Bi_2 , against the VASP reference 4.097. This near-agreement is included as a diagnostic and is not treated as a validated workflow result.

electrons assumed occupied	Ba_3BiSb	Ba_3Bi_2
56	2.12	2.10
58	1.43	1.44
60 (fixed count used here)	2.02	2.78
62 (<code>jarvis-tools</code> , occupation ≥ 0.5)	0.08	4.12
64	1.63	4.22
VASP reference	2.267	4.097

Implications. For a clean-gap insulator, the occupied-state count is fixed by the electronic structure, and the present workflow reproduces the published VASP value within 0.9%. For a zero-gap semimetal, the convention used to define the occupied electronic states can dominate the result. The 11% discrepancy for Ba_3BiSb is plausibly dominated by this ambiguity rather than by an incorrectly inferred general QE–VASP difference. This interpretation is supported by the close agreement generally found between modern plane-wave codes for ground-state quantities [26] and by the 0.9% agreement obtained here for insulators using the same computational procedure. The different $\mathbf{k}$ -point meshes and material-specific pseudopotential effects remain confounding factors.

This interpretation is not definitive because none of the occupation counts considered above reproduces the reference value of 2.267. The closest result is 2.12 for 56 electrons. The analysis establishes that occupation-count ambiguity is sufficiently large to account for the observed discrepancy, but it does not necessarily establish causality. A definitive assessment would require mesh-converged calculations and the VASP occupation and degeneracy treatment for the same structure. What is definite is that both the published VASP result and the fixed-60 QE result support a high-spillage classification for Ba_3BiSb , although the numerical value remains sensitive to the occupied-band convention.

PbTe. The spillage peaks at the four equivalent L points, where each value is 2.0103, and decreases to approximately 0.09 at neighbouring points. PbTe has normal band ordering at L and is topologically trivial at ambient pressure, in contrast to the band-inverted topological crystalline insulator SnTe [27]. The large spillage therefore indicates a strong SOC-induced change in the occupied electronic states relative to the scalar-relativistic reference, but it does not establish a band inversion or non-trivial topology. This distinction illustrates why a high-spillage candidate requires subsequent evaluation of topological invariants or surface states. Choudhary et al. [2] likewise report rare high-spillage false positives. Independent confirmation of the numerical value remains necessary.

A recent paper predicts a spillage value of 2.200 for PbTe [28], a 9.5% difference from the present value. Other reported values are 2.320 for BaMg_2Bi_2 (11.8%), 2.315 for Bi_2Te_3 (10.6%), and 2.306 for Ba_3BiSb (1.7%) [28]. These differences, from 1.7% to 11.8%, are comparable to those found among the spillage values computed here.

13 Computational reproducibility

The input decks, post-processing code, reference calculations, and this document are organized in the standalone `qe-spillage/` repository. All commands below are run from the repository root. The QE input decks are in `inputs/`, the post-processors are at the repository root, and the recorded SCF logs and spillage tables are under `reference/`. QE resolves `pseudo_dir` and `outdir` relative to the repository root. The untracked pseudopotentials are placed in `pseudo_nc/`, while the two generated `outdir` paths are `out_nosoc_nc_BaMg2Bi2/` and `out_soc_nc_BaMg2Bi2/`, as shown in Listing 4. The environment variable `OMP_NUM_THREADS=1` prevents oversubscription when eight MPI processes are used. The `-npool 8` option assigns one MPI process to each of eight **k**-point pools. These parallelization settings enhance computational performance but do not affect the calculated wavefunctions or spillage.

The complete end-to-end command sequence used for the representative calculation is given in Listing 4.

Listing 4: End-to-end commands for the worked example of BaMg2Bi2.

```
# 1. Install QE (conda-forge build runs under Rosetta on Apple Silicon)
CONDA_SUBDIR=osx-64 conda create -n qe -c conda-forge qe -y

# 2. Put PseudoDojo NC v0.4 PBE SR/FR files in pseudo_nc/

# 3. From the qe-spillage repository root, run the two SCF calculations
conda activate qe
export OMP_NUM_THREADS=1
mpirun -np 8 pw.x -npool 8 -in inputs/BaMg2Bi2.scf.nosoc.nc.in > nosoc_nc_BaMg2Bi2.out
# ~2 min
mpirun -np 8 pw.x -npool 8 -in inputs/BaMg2Bi2.scf.soc.nc.in > soc_nc_BaMg2Bi2.out
# ~13 min

# 4. Compute the spillage from the two wavefunction sets
SPILLAGE_NELEC=60 SPILLAGE_REFERENCE=2.075 python compute_spillage.py \
    out_nosoc_nc_BaMg2Bi2/BaMg2Bi2_nosoc_nc.save \
    out_soc_nc_BaMg2Bi2/BaMg2Bi2_soc_nc.save
# -> max gamma = 2.0941 at Gamma
```

The files `inputs/BaMg2Bi2.scf.nosoc.nc.in` and `inputs/BaMg2Bi2.scf.soc.nc.in` contain the two QE inputs. The root-level script `compute_spillage.py` performs the post-processing. The directory `reference/scf-logs/` contains the saved QE outputs, while `reference/spillage/` contains the recorded per-**k** spillage tables for all materials. The `pseudo_nc/` and generated `out_*/` directories are intentionally excluded from version control. The LaTeX source and rendered document are in `docs/`.

14 Relation to prior work

This section distinguishes established methods from the contributions of the present work.

Established methods. Spin-orbit spillage was introduced by Liu and Vanderbilt [1], and Equation (1) corresponds to their equation 4. The use of spillage as a high-throughput screen for topological materials, including the $\gamma > 0.5$ rule of thumb and the screening of the JARVIS database, is due to Choudhary and co-workers [2, 3, 4], and the machine-learning classifier that motivates this

work is from Ref. [4]. The BaMg_2Bi_2 and Ba_3BiSb structures and their reference spillages are from the JARVIS database [2, 20]. The underlying methods are established, and these include Kohn–Sham DFT [8, 7], the PBE functional [9], Quantum ESPRESSO [5, 6], the Dirac equation [10] and its scalar-relativistic reduction [11], relativistic and optimized norm-conserving pseudopotentials [13, 12, 15, 16], SOC in a plane-wave pseudopotential code [14], PAW [18], Monkhorst–Pack $\mathbf{k}$ -grids [21], Methfessel–Paxton smearing [22], and Löwdin orthonormalization [23].

Brillouin-zone sampling in previous work. Ref. [1] specifies the use of QE, PBE, norm-conserving pseudopotentials from the OPIUM package, and cutoffs of 55 Ry for Bi_2Se_3 and Sb_2Se_3 and 65 Ry for In_2Se_3 , “with an $8 \times 8 \times 8$ Monkhorst–Pack $\mathbf{k}$ mesh,” from which the plane-wave wavefunctions are extracted and their equation 6 is used to evaluate the spillage. Thus, both studies evaluate the DFT wavefunctions on a regular $\mathbf{k}$ -point mesh, although the computational control parameters differ.

High-symmetry lines are used for visualization rather than Brillouin-zone sampling. Liu and Vanderbilt used charts and maps to show the momentum dependence of $\gamma(\mathbf{k})$. Their Figure 5(a) traces γ along a path through the zone, and Fig. 5(b) shows the spillage as a colour map over the (k_x, k_y) plane of Bi_2Se_3 at $k_z = 0$. The high-symmetry path includes Γ , where the inversion is expected. Section II B of Ref. [1] shows that a topologically non-trivial system must have $\gamma(\mathbf{k}) \geq 1$ somewhere in the zone, and for an inversion-symmetric $\mathbb{Z}_2$ insulator the two partner inversions at $\mathbf{k}_0$ and $-\mathbf{k}_0$ merge at a TRIM, so one expects $\gamma \geq 2$ there.

The high-throughput screen of Choudhary and co-workers [2, 4] reduces each material to $\max_{\mathbf{k}} \gamma(\mathbf{k})$ for use as a thresholded machine-learning label. The accuracy of our maximum depends on the sampled $\mathbf{k}$ -point set, motivating the mesh-convergence considerations in Section 10.

Contribution of the present work. The reference implementation in `jarvis-tools` and the high-throughput spillage screens built on it [2, 3, 4], obtain wavefunctions from VASP [19] by reading `WAVECAR` files. Computational reproduction of that workflow therefore requires access to VASP, which is expensive.

QE has previously been used in related stages of the spillage workflow. Liu and Vanderbilt used QE for their original spillage calculations, as described above. Ref. [2] also used QE for Wannier interpolation and evaluation of topological invariants after screening 289 candidate materials, but the spillage itself was calculated with VASP.

No reusable QE-based implementation of the spillage calculation was reported, and no code accompanies Ref. [1]. In our present contribution, the post-processor described in Section 9 reads QE’s per- $\mathbf{k}$ -point HDF5 wavefunction files and enables the calculation without VASP.

Liu and Vanderbilt report a spillage of 2.12 at Γ for Bi_2Se_3 , computed in QE with norm-conserving pseudopotentials. The present calculation for Bi_2Se_3 (JVASP-1067), using a $6 \times 6 \times 6$ mesh, 78 electrons, and PseudoDojo NC pseudopotentials, gives 2.1146 and is also peaked at Γ . The difference is 0.3% despite the use of different pseudopotential types, cutoffs, and meshes, providing direct validation against the work that introduced the quantity.

First, additional results support this agreement. The related compound Bi_2Te_3 (JVASP-25) gives 2.1116 compared with its VASP reference of 2.094. Both compounds peak at Γ . Their second-largest sampled values are 0.9851 and 1.8294 for Bi_2Se_3 and Bi_2Te_3 , respectively. The latter is only 13% below the sampled maximum, so these calculations do not establish formal $\mathbf{k}$ -mesh convergence. A refined mesh retains the value at Γ but could identify a larger value at an intermediate point. Repeating both calculations with `occupations='fixed'` confirms the values to four decimal places.

The band-gap comparison is less uniform than the spillage comparison. The calculated non-SOC direct gaps agree with JARVIS, giving 0.279 versus 0.279 eV for Bi_2Se_3 and 0.348 versus 0.342 eV for Bi_2Te_3 . In contrast, the calculated SOC direct gaps are 0.273 and 0.292 eV for Bi_2Se_3 and Bi_2Te_3 , respectively, compared with the JARVIS values of 0.186 and 0.223 eV. The $6 \times 6 \times 6$ mesh likely misses an off- Γ SOC gap minimum. This gap discrepancy does not directly alter the reported spillage at Γ , which is sampled by both meshes, but it reinforces the need for an explicit mesh-convergence study.

The Bi_2Se_3 spillage agrees within one percent with both the QE value of 2.12 reported by Liu and Vanderbilt [1] and the JARVIS VASP value of 2.098 [2]. Together with the systematic offset discussed in Section 12, these results indicate that the QE implementation reproduces published spillage values for systems with well-defined occupied subspaces, subject to the stated mesh limitation.

The pseudopotential choice in Section 7 has a methodological consequence. With PAW pseudopotentials the true overlap requires the augmentation operator S , which is not stored in the wavefunction file. After DFT, the `jarvis-tools` implementation handles this augmentation by orthogonalizing the occupied coefficient matrices numerically via SVD, under the assumption that the SOC and non-SOC wavefunctions span the same space. This orthogonalization is an uncontrolled approximation and changed the FR-PAW result from a raw $\gamma \approx -17$ to 1.63. Choosing *norm-conserving* pseudopotentials removes the need for PAW augmentation in this overlap: the coefficient dot product in Equation (6) is then the appropriate pseudo-wavefunction overlap. Consequently, agreement of `gamma_raw` and `gamma_lowdin` is expected for NC wavefunctions and does not validate the metric independently. Their agreement at every $\mathbf{k}$ -point is retained as a regression test for plane-wave alignment and basis consistency. Liu and Vanderbilt also used norm-conserving pseudopotentials, so their Eq. (6) likewise uses the ordinary coefficient overlap.

The underlying principle is established. Klimeš, Kaltak, and Kresse [17] show, in the context of *GW* quasiparticle energies, that PAW describes overlap integrals incorrectly because the partial-wave basis inside the atomic spheres is incomplete, and that the error “can be avoided by adopting norm-conserving partial waves.” The same mechanism applies here to the spillage overlap. The present work shows that the pseudopotential type determines the appropriate overlap metric in Eq. (1). The `gamma_raw/gamma_lowdin` comparison provides a regression test of the implementation rather than independent evidence for that metric. To the best of our knowledge, the connection between pseudopotential type and the spillage overlap metric has not been stated explicitly in the spillage literature.

The numerical results obtained in the present work are $\eta = 2.094$ for BaMg_2Bi_2 compared with the 2.075 reference, 2.016 and 1.632 for Ba_3BiSb using NC and PAW pseudopotentials, respectively, and 2.010 for PbTe , compared with a reported value of 2.200 [28]. The analysis in Section 12 demonstrates that, for a semimetal with a degenerate manifold at the Fermi level, the spillage is ill-conditioned with respect to the occupied-band count and varies between 0.08 and 4.2 across plausible choices. This sensitivity is a general limitation of the method rather than one specific to the present implementation.

The two Ba_3BiSb runs also changed the cutoff, $\mathbf{k}$ -mesh and valence-electron count, so the difference between the corrected values 1.632 and 2.016 cannot be assigned to the pseudopotential alone. The raw PAW result near -17 nevertheless establishes that an ordinary coefficient dot product does not provide the correct PAW overlap in the absence of the augmentation metric.

Author contributions and AI-assisted editing

Duy Quan Nguyen carried out the calculations, implemented the reproducibility workflow, analyzed the results, and prepared the initial manuscript. Paul C. H. Li supervised the project, contributed to the interpretation of the results, and reviewed and edited the manuscript. Both authors approved the final manuscript.

AI-assisted tools were used solely for language and L^AT_EX editing. All scientific content, calculations, analysis, and conclusions were produced and verified by the authors.

Glossary

DFT / Kohn–Sham (K–S)

Density functional theory (DFT) [7], a widely used method for calculating electronic properties (wavefunctions and density) of solids. DFT maps the interacting many-electron problem onto a set of effective single-electron Kohn–Sham (K–S) equations.

SCF

The self-consistent-field (SCF) loop is the iterate-until-stable loop QE uses to solve the K–S equations in DFT to self-consistency or convergence.

Spin–orbit coupling (SOC)

A relativistic effect coupling the spin to the orbital angular momentum, a magnetic effect, of an electron, Equation (4). Strong in heavy atoms (Bi, Pb). SOC is what can invert bands and create topological materials.

Dirac equation

The relativistic wave equation for an electron [10]. Spin–orbit coupling drops out of it as a low-velocity correction; it is solved per atom when a fully-relativistic pseudopotential is generated.

Scalar- vs fully-relativistic (SR / FR)

SR keeps the relativistic energy corrections but averages spin–orbit coupling away; FR keeps it, storing separate $j = \ell \pm \frac{1}{2}$ channels. The two calculations use `_SR.upf` and `_FR.upf` files, respectively. See Section 6.

Spinor

A wavefunction with a spin-up and a spin-down component. SOC calculations use spinors.

Band inversion

When SOC swaps the character of an occupied and an empty band. It is an important indicator of possible topological character, not by itself a proof, and is what large spillage is designed to detect.

Spillage $\gamma(\mathbf{k})$, η

Equations (1) and (6): how much the filled electronic states change when SOC is turned on. $\eta = \max_{\mathbf{k}} \gamma$ is the reported per-material spillage value.

Pseudopotential

An effective potential replacing the nucleus and core electrons, so only valence electrons are computed. NC and PAW are two types of pseudopotentials (PP); SR and FR PP are without and with SOC, respectively.

Plane-wave cutoff (`ecutwfc`, `ecutrho`)

Energy limits setting how many plane waves represent the electron wavefunctions (wfc) and density

(rho). A higher cutoff is more accurate but more costly. Note these are energies in Rydberg, not counts: `ecutwfc = 70` keeps every $\mathbf{G}$ with $|\mathbf{k} + \mathbf{G}|^2 \leq 70$, which for BaMg₂Bi₂ has a count of 11,573 plane waves.

G vectors

The reciprocal-lattice vectors of the crystal, and the frequencies of the Fourier series in which every wavefunction is expanded. Which ones are present is decided by the cutoff (a sphere in reciprocal space) and listed in the wavefunction file as `MillerIndices`, the integer triples (h, k, l) that label each one. The identical cell, cutoff and matched $\mathbf{k}$ -points give the same plane-wave basis in the two runs; the identical crystal structure and `nosym/noinv` ensure that QE writes the same full $\mathbf{k}$ -point list rather than reducing it differently. See Eq. (5) for counting the $\mathbf{G}$ -vectors.

Plane-wave coefficients

The list of complex numbers $c_{n\mathbf{k}}(\mathbf{G})$ multiplying each plane wave; see Equation (3). A single band at a single $\mathbf{k}$ -point is this list, stored in the wavefunction file as `evc`. With norm-conserving pseudopotentials, the overlap $\langle \psi | \psi' \rangle$ of Eq. (1) is the dot product of two such lists. In the SOC run each band carries two of these lists back to back, one per spin component.

FFT grid

The uniform real-space and reciprocal-space grids that QE transforms between during each SCF iteration. QE distinguishes smooth and dense grids. The density cutoff `ecutrho` controls the dense grid, reported here as $50 \times 50 \times 90$. The cutoffs are therefore parameters of the Fourier representation.

Monkhorst–Pack mesh

The regular grid of $\mathbf{k}$ -points used to sample the Brillouin zone [21]. The mesh used here is $6 \times 6 \times 4$, giving 144 points. This grid is distinct from a high-symmetry *path*, which is a line through selected points used for plotting band structures and, in Ref. [1], for plotting $\gamma(\mathbf{k})$.

Occupation numbers f

The per-band fillings QE writes under `occupation numbers` in its output, on a 0–1 scale in both calculations. For an ideal insulator, f would be exactly 1 below the gap and 0 above it. With Methfessel–Paxton (MP) smearing it is not: the scheme can overshoot above 1 and go slightly negative by construction. If the smearing width exceeds the gap, the charge is also transferred across the gap. See Table 4. The post-processor does not read these values and instead uses a fixed N_{occ} justified by the band energies.

MPI / $\mathbf{k}$ -point pool

MPI (Message Passing Interface) is the protocol that lets several processes work on one DFT calculation. A pool is a group of processes assigned its own subset of $\mathbf{k}$ -points (`-npool`). This pool arrangement can work well when there are many $\mathbf{k}$ -points, but the best pool count must be timed. Pool parallelism does not distribute the main real- and reciprocal-space arrays, so increasing the number of pools can increase memory use. The pool changes the speed and memory layout, never the physical answer.

SVD (singular value decomposition)

SVD is a factorisation of a matrix into two unitary matrices (V, W) and a non-negative diagonal matrix Σ of singular values. Liu and Vanderbilt apply SVD in their valence-band-resolved spillage analysis to the overlap matrix $L_{nm}(\mathbf{k}) = \langle \psi_{n\mathbf{k}} | \tilde{\psi}_{m\mathbf{k}} \rangle$ connecting occupied states *without* SOC to *unoccupied* states *with* SOC. Writing $L = V\Sigma W^\dagger$ and transforming the two sets by V and W makes the overlap matrix between them real and diagonal, which is the gauge in which the band-resolved spillage is read off. Note this is a different matrix from the occupied-occupied overlap

$M_{mn}(\mathbf{k}) = \langle \psi_{m\mathbf{k}} | \tilde{\psi}_{n\mathbf{k}} \rangle$ of their equation 5 [1], which is the one Eq. (1) above is built from and to which no SVD is applied. Separately, `jarvis-tools` uses SVD to orthogonalise PAW coefficient matrices numerically, also as a stand-in for the augmentation operator S . That second use is an approximation. With norm-conserving pseudopotentials it is unnecessary, and the related Löwdin transformation changes no values at the reported precision of four decimal places.

k-point / Brillouin zone

The Brillouin zone is the crystal’s momentum space, which is the reciprocal space of real space; $\mathbf{k}$ -points are the sample grid over reciprocal space. $\Gamma = (0, 0, 0)$ is its center.

TRIM points

The time-reversal-invariant momenta (TRIM): high-symmetry $\mathbf{k}$ -points (Γ , and zone-boundary points) where band inversions often occur. For an inversion-symmetric insulator, the Fu–Kane parity criterion evaluates the $\mathbb{Z}_2$ indices from parity eigenvalues at the TRIM [25]; more general systems require other formulations.

Semimetal

A material whose occupied and empty bands overlap in energy, so there is no gap or zero band gap. The spillage is not uniquely defined; see Section 12.

Smearing width (degauss)

A small numerical softening of the filled/empty boundary, needed for metals so the SCF is stable. For an insulator the width should be narrower than the band gap. Ours is Methfessel–Paxton [22] at 0.01 Ry = 0.136 eV, which is narrower than the non-SOC gap but wider than the SOC one, hence Table 4.

nosym / noinv

Turn off symmetry reduction and time-reversal folding, so the two SCF runs share an identical basis of $\mathbf{k}$ -points and plane waves. The SOC and non-SOC wavefunctions can then be overlapped and compared to compute the spillage.

Löwdin orthonormalization

A standard way to force a set of states to be mutually orthonormal [23]. The orthonormalization was used to approximate the missing PAW augmentation. With NC pseudopotentials this is unnecessary.

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